

RETAILMART V3 • SQL

Scalar Functions — Quick Quiz

Answer Before Looking!

For each question, try to answer BEFORE scrolling to the answer. This builds interview confidence. Best used a day or two AFTER the Scalar Functions class.

Quiz 1: What does this return?

EASY

ANSWER: 'Hello World Sql' — INITCAP capitalizes the first letter of EVERY word and lowercases the rest.

Question

 ANSWER

```
SELECT INITCAP('hello WORLD sql');
```

Quiz 2: What does this return?

EASY

ANSWER: NULL — the `||` operator propagates NULL. If ANY part is NULL, the entire result is NULL. Use `CONCAT()` instead for NULL-safety.

Question

 ANSWER

```
SELECT 'Hello' || NULL || 'World';
```

Quiz 3: What does this return?

MEDIUM

ANSWER: 0 — May 18, 2025 is a Sunday. DOW returns 0 for Sunday, 1 for Monday ... 6 for Saturday.

Question



ANSWER

```
SELECT EXTRACT(DOW FROM DATE '2025-05-18');
```

Quiz 4: What does this return?

MEDIUM

ANSWER: '2025-07-01 00:00:00' — DATE_TRUNC rounds DOWN to the first moment of the month. Time is set to midnight.

Question



ANSWER

```
SELECT DATE_TRUNC('month', TIMESTAMP  
    '2025-07-15 14:30:00');
```

Quiz 5: Spot the Bug

HARD

ANSWER: If phone is NULL, LENGTH(NULL) = NULL, and NULL = 10 is UNKNOWN -> row is skipped silently. Fix: add WHERE phone IS NOT NULL AND LENGTH(phone) = 10. Also, the || concatenation will return NULL if last_name is NULL — use CONCAT instead.

Question

 ANSWER

```
SELECT first_name || ' ' || last_name,  
       LENGTH(phone),  
       SPLIT_PART(email, '@', 2)  
FROM customers.customers  
WHERE LENGTH(phone) = 10  
LIMIT 10;
```

Quiz 6: Which is Faster?

HARD

ANSWER: Option B is faster! It can use an index on `order_date` directly. Option A calls `EXTRACT` on every row first (function on column = no index use). Always prefer range comparisons over function-based filtering when possible.

Option A



ANSWER

```
WHERE EXTRACT(YEAR FROM order_date) = 2025  
      AND EXTRACT(MONTH FROM order_date) = 3
```

Option B



ANSWER

```
WHERE order_date >= '2025-03-01'  
      AND order_date < '2025-04-01'
```